

Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.77)

Metric

$$ds^2 = -2 H du \otimes du - du \otimes dr - dr \otimes du + \frac{2 r^2}{P^2} dx \otimes dx + \frac{2 r^2}{P^2} dy \otimes dy$$

Metric Functions

$$H(u, r, x, y) = H$$

$$P(u, x, y) = P$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Null Tetrad

Covariant components

$$l = (1) \, du$$

$$n = (H) \, du + (1) \, dr$$

$$m = \left(\frac{r}{P}\right) dx + \left(\frac{i r}{P}\right) dy$$

$$\bar{m} = \left(\frac{r}{P}\right) dx + \left(-\frac{i r}{P}\right) dy$$

Contravariant components

$$l = (-1) \, dr$$

$$n = (-1) \, du + (H) \, dr$$

$$m = \left(\frac{P}{2r} \right) dx + \left(\frac{iP}{2r} \right) dy$$

$$\overline{m} = \left(\frac{P}{2r} \right) dx + \left(-\frac{iP}{2r} \right) dy$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u + H \, \partial_r$$

$$\delta = \frac{P}{2r} \partial_x + \frac{iP}{2r} \partial_y$$

$$\bar{\delta} = \frac{P}{2r} \partial_x - \frac{iP}{2r} \partial_y$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{HP + r \frac{\partial}{\partial u} P}{rP}$$

$$\lambda = 0$$

$$\nu = \frac{\left(\frac{\partial}{\partial x} H - i \frac{\partial}{\partial y} H \right) P}{2r}$$

$$\alpha = \frac{\frac{\partial}{\partial x} P - i \frac{\partial}{\partial y} P}{4r}$$

$$\beta = -\frac{\frac{\partial}{\partial x} P + i \frac{\partial}{\partial y} P}{4r}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{1}{2} \frac{\partial}{\partial r} H$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{2r^2P\frac{\partial^2}{(\partial r)^2}H - P\frac{\partial}{\partial x}P^2 + P^2\frac{\partial^2}{(\partial x)^2}P - P\frac{\partial}{\partial y}P^2 + P^2\frac{\partial^2}{(\partial y)^2}P - 4HP - 4r\frac{\partial}{\partial u}P}{8r^2P}$$

$$\Phi_{12} = -\frac{P^2\frac{\partial^2}{\partial r\partial x}H + iP^2\frac{\partial^2}{\partial r\partial y}H + P\frac{\partial^2}{\partial u\partial x}P + iP\frac{\partial^2}{\partial u\partial y}P - \frac{\partial}{\partial u}P\frac{\partial}{\partial x}P - i\frac{\partial}{\partial u}P\frac{\partial}{\partial y}P}{4rP}$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = -\frac{P^2\frac{\partial^2}{\partial r\partial x}H - iP^2\frac{\partial^2}{\partial r\partial y}H + P\frac{\partial^2}{\partial u\partial x}P - iP\frac{\partial^2}{\partial u\partial y}P - \frac{\partial}{\partial u}P\frac{\partial}{\partial x}P + i\frac{\partial}{\partial u}P\frac{\partial}{\partial y}P}{4rP}$$

$$\Phi_{22} = \frac{P^4\frac{\partial^2}{(\partial x)^2}H + P^4\frac{\partial^2}{(\partial y)^2}H + 4r^2P\frac{\partial}{\partial r}H\frac{\partial}{\partial u}P + 4rP^2\frac{\partial}{\partial u}H - 8rHP\frac{\partial}{\partial u}P - 8r^2\frac{\partial}{\partial u}P^2 + 4r^2P\frac{\partial^2}{(\partial u)^2}P}{4r^2P^2}$$

$$\Lambda = -\frac{2r^2P\frac{\partial^2}{(\partial r)^2}H + 8rP\frac{\partial}{\partial r}H + P\frac{\partial}{\partial x}P^2 - P^2\frac{\partial^2}{(\partial x)^2}P + P\frac{\partial}{\partial y}P^2 - P^2\frac{\partial^2}{(\partial y)^2}P + 4HP + 12r\frac{\partial}{\partial u}P}{24r^2P}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{2r^2\frac{\partial^2}{(\partial r)^2}H - 4r\frac{\partial}{\partial r}H + \frac{\partial}{\partial x}P^2 - P\frac{\partial^2}{(\partial x)^2}P + \frac{\partial}{\partial y}P^2 - P\frac{\partial^2}{(\partial y)^2}P + 4H}{12r^2}$$

$$\Psi_3 = -\frac{rP^2\frac{\partial^2}{\partial r\partial x}H - irP^2\frac{\partial^2}{\partial r\partial y}H - 2P^2\frac{\partial}{\partial x}H + 2iP^2\frac{\partial}{\partial y}H - rP\frac{\partial^2}{\partial u\partial x}P + irP\frac{\partial^2}{\partial u\partial y}P + r\frac{\partial}{\partial u}P\frac{\partial}{\partial x}P - ir\frac{\partial}{\partial u}P\frac{\partial}{\partial y}P}{4r^2P}$$

$$\Psi_4 = \frac{\left(P\frac{\partial^2}{(\partial x)^2}H - 2iP\frac{\partial^2}{\partial x\partial y}H - P\frac{\partial^2}{(\partial y)^2}H + 2\frac{\partial}{\partial x}H\frac{\partial}{\partial x}P - 2i\frac{\partial}{\partial y}H\frac{\partial}{\partial x}P - 2i\frac{\partial}{\partial x}H\frac{\partial}{\partial y}P - 2\frac{\partial}{\partial y}H\frac{\partial}{\partial y}P\right)P}{4r^2}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"